Checkpoint 1 — AGI: On-the-Spot Reasoning

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Data: The datasets we plan to use are publicly available

90 minutes working offline as a team: yes

Motivation and research question

This project is based on "A Definition of AGI" (Hendrycks et al., 2025), which gives a quantifiable framework to analyse whether an AI model can be defined to be AGI. It comprises ten different cognitive components, and our project will explore On-the-Spot Reasoning. Hendrycks et al. (2025) divide this into deduction (20%), induction (40%), theory of mind (20%), planning (10%) and adaptation (10%). The paper applied their framework to GPT-4 and GPT-5: GPT-4 got 0 out of 10 points for On-the-Spot Reasoning, while GPT-5 got 7 out of 10. We aim to apply their framework, specifically On-the-Spot Reasoning, to a new flagship model, Claude Opus 5.5..

We have to approach the benchmarks that the authors used critically when we apply them to Claude Opus 5.5. Some of them were already saturated by GPT-5, and newer models might have seen these older benchmarks during training, so high scores might partly reflect memorisation. When Mirzadeh et al. (2025) changed only the names and numbers in school maths problems, model accuracy dropped and varied a lot between versions, even though the reasoning needed to stay the same. Similarly, M. K. Chen et al. (2025) introduced JustLogic, a synthetically generated deductive reasoning benchmark designed to reduce the influence of prior knowledge; on it, the best reasoning models only matched the average human. Together, these studies motivate us to evaluate both performance on the original benchmarks and robustness on additional or modified tasks.

Research question: Can Claude Opus 5.5 reach human level performance in the five reasoning categories?

Hypothesis 1: We expect Claude to fall below human level on at least some tests

Hypothesis 2: Performance will decrease on at least some modified or additional benchmark tasks compared with the original tasks, indicating that high scores on standard benchmarks may not fully reflect robust reasoning.

Methods:

We plan to use Claude Opus 5.5 to run the following benchmarks on medium reasoning. Each of us will be responsible for a category of On-the-Spot reasoning domain.

1. Deduction 20% - Marten Mathias Jaani

Deduction is the process of reasoning from one or more premises to reach a conclusion that is logically guaranteed to be true. Hendrycks et al. (2025) use 86% accuracy on LogiQA 2.0 (Liu et al., 2023) as the human-level threshold for this category, subject to memorisation and robustness checks. As an additional benchmark we use JustLogic (M. K. Chen et al., 2025): its authors argue that LogiQA-style tests have complex language but simple arguments, letting models rely on prior knowledge, whereas JustLogic's synthetic items isolate deduction.

2. Induction 40% - Alexandru Savu

Induction is the capacity to determine general rules and principles underlying a problem or phenomenon, based on specific observations. The ARC-AGI-2 (Chollet et al., 2025) public training and evaluation datasets will be used to test the model, and the ARC-AGI-2 Human testing dataset will be used for establishing a human baseline. If the model performs worse than the average, the score does not increase. If it performs above average but beneath the 90th percentile, the score will increase by 20%. A score above the 90th percentile corresponds to a full 40% increase in the On-the-Spot reasoning score. Additional benchmarks are taken from the survey by K. Chen et al. (2026).

3. Theory of Mind 20% - Aaron Anders Jaani

Theory of Mind means understanding what other people know or believe. We will use two public benchmarks, ToMBench (Z. Chen et al., 2024) and FANToM (Kim et al., 2023). We will test 400 ToMBench questions (50 from each task) and 100 FANToM question sets. We will compare the results with the published human scores (85.4% and 87.5%), using average accuracy across the eight ToMBench tasks and the All score for FANToM. If the usage limits allow it, we will run more questions. As an additional benchmark: OpenToM (Xu et al., 2024) tests how well models understand characters' beliefs and attitudes in short stories. We can test if Claude performs differently on these types of questions

4. Planning 10% - Hannes Arumäe

Planning is the ability to devise a sequence of actions that leads from an initial state to a desired goal state. Following Hendrycks et al. (2025), the model must reach at least 90% accuracy on both Natural Plan (Zheng et al., 2024) and PlanBench Blocksworld (Valmeekam et al., 2023). Natural Plan contains trip-planning, meeting-planning and calendar-scheduling tasks written in natural language. Blocksworld asks the model to restack blocks to reach a goal configuration. We will run 100 Natural Plan questions per task (300 in total) and all 600 Blocksworld plan-generation instances. Each answer is checked automatically against the correct solution. As a contamination check, we will also run 100 "Mystery Blocksworld" instances, where the action names are replaced with unrelated words. As an additional benchmark we use TravelPlanner (Xie et al., 2024), multi-day trips with many constraints (budget, accommodation, restaurants); at publication GPT-4 solved only 0.6%, so it is far from saturated.

5. Adaptation 10% - Martin Linna

Adaptation is the power to find a new pattern according to basic feedback, and to drop it when the rule changes without warning. We will measure it with the 64-card Wisconsin Card Sorting Test (WCST; Grant & Berg, 1948), implemented as in Hao et al. (2025): four key cards differ in colour, shape and number, and on each of 64 trials Claude Opus 5.5 sees one response card and answers with a pile number (1-4). The program replies only "Correct" or "Incorrect", and the hidden rule changes after 10 consecutive correct answers. The standard from Hendrycks et al. (2025) is fewer than 15 Total Errors. We will run two conditions, 20-30 independent runs each, with a new random card order per run: the standard WCST, and a relabelled version with new attributes, new wording and a shuffled rule order, to check for memorization.

Analysis

We run each benchmark and compare Claude's results with the human baselines. Points earned in the five categories add up to an On-the-Spot Reasoning score from 0 to 10. We also compare results on the original and modified or additional tasks to test hypothesis 2.

Weekly plan until Checkpoint 2:

20–26 Oct: Read more relevant papers and agree on the tests.

27 Oct–2 Nov: Meet with the team, set up the Claude Code scripts and do small test runs.

3–9 Nov: Benchmarks outlined in the framework are completed.

10–16 Nov: Compare the results with human scores and prepare Checkpoint 2.

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